ADVENTUREWORKS: END-TO-END SALES ANALYTICS SOLUTION REPORT

GITHUB LINK: https://github.com/sangvivy/adventureworks-sales-intelligence

Part A: Data Acquisition and Understanding

1. Domain and Business Problem

Domain: Retail & E-commerce Sales Analytics

Business Problem: AdventureWorks, a global manufacturing company, operates in a highly competitive market selling cycling equipment and accessories. Currently, their data is siloed across multiple years and categories, making it difficult for leadership to track year-over-year growth, identify high churn customer segments, or pinpoint regional underperformance.

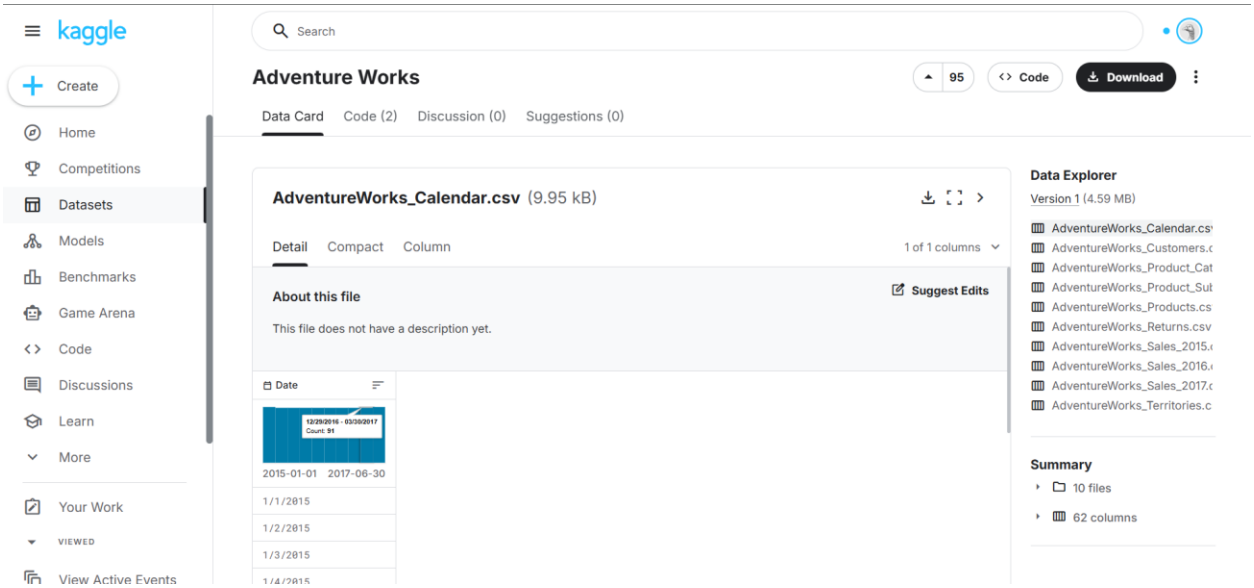
The goal of this Power BI solution is to design a **Strategic Executive Dashboard** that transforms raw transactional data into actionable insights regarding revenue trends, product profitability, and regional sales efficiency to support data-driven decision-making.

2. Dataset Source

https://www.kaggle.com/datasets/ukveteran/adventure-works?resource

The dataset is the **AdventureWorks Business Intelligence Suite**, a multi-relational dataset containing historical sales, product metadata, customer demographics, and territorial mappings.

- **Source Format:** 10+ Related CSV (Comma Separated Values) files.
- **Temporal Scope:** Data spans from 2015 to 2017.



3. Key Tables and Fields

The solution utilizes a **Star Schema** architecture involving the following primary tables:

Table Name	Type	Key Fields	Variables Included
AW_Sales (2015-2017)	**Fact**	OrderDate, ProductKey, CustomerKey, TerritoryKey, OrderQuantity	**Numerical:** Quantities, Order IDs. This is the transactional core.
AW_Products	**Dim**	ProductKey, ProductName, ProductCost, ProductPrice	**Categorical:** Names, Models. **Numerical:** Pricing and Costs for Margin analysis.
AW_Customers	**Dim**	CustomerKey, Gender, EducationLevel, Occupation	**Categorical:** High-level demographic segments for customer profiling.
AW_Calendar	**Dim**	Date, MonthName, Quarter, Year	**Date:** Essential for Time-Intelligence and Trend analysis.
AW_Territories	**Dim**	SalesTerritoryKey, Region, Country, Continent	**Categorical/Geospatial:** Used for regional performance mapping.
AW_Returns	**Fact**	ProductKey, ReturnDate,ReturnQuantity,TeritoryKey	**Date:** ReturnDate. **Numerical:** ProductKey,Quantity,TerritoryKey

4. Suitability for Advanced Analysis

I selected this dataset specifically because it meets the requirements.

- **Data Volume:** The combined Sales tables (2015, 2016, and 2017) contain over **56,000 rows**, providing a "Big Data" feel that tests model performance and DAX efficiency.
- **Relational Complexity:** The data is spread across **10 separate files**. This requires advanced Power Query skills (Appending sales years) and complex Data Modeling (establishing One-to-Many relationships between Sales and Dimensions).
- **Analytical Depth: * Time Intelligence:** The inclusion of a dedicated Calendar table allows for advanced DAX measures like Total Revenue YTD, Previous Month Sales, and YoY Growth %.
    - **Profitability Modeling:** By relating ProductCost from the Products table to OrderQuantity in the Sales table, we can calculate dynamic **Gross Profit Margins**, moving beyond simple sales tracking into financial intelligence.
- **Categorical Diversity:** With fields like Gender, Education, and Occupation, the dataset allows for **Advanced Customer Segmentation** and "What-If" parameter analysis.

Raw Data Source Files

Name	Type	Compressed size	Password p...	Size	Ratio	Date modified
AdventureWorks_Calendar	Microsoft Excel Comma S...	2 KB	No	10 KB	82%	1/3/2020 10:38 PM
AdventureWorks_Customers	Microsoft Excel Comma S...	384 KB	No	1,918 KB	81%	1/3/2020 10:38 PM
AdventureWorks_Product_Categories	Microsoft Excel Comma S...	1 KB	No	1 KB	3%	1/3/2020 10:38 PM
AdventureWorks_Product_Subcateg...	Microsoft Excel Comma S...	1 KB	No	1 KB	48%	1/3/2020 10:38 PM
AdventureWorks_Products	Microsoft Excel Comma S...	10 KB	No	57 KB	83%	1/3/2020 10:38 PM
AdventureWorks_Returns	Microsoft Excel Comma S...	7 KB	No	34 KB	81%	1/3/2020 10:38 PM
AdventureWorks_Sales_2015	Microsoft Excel Comma S...	27 KB	No	116 KB	77%	1/3/2020 10:38 PM
AdventureWorks_Sales_2016	Microsoft Excel Comma S...	222 KB	No	1,059 KB	80%	1/3/2020 10:38 PM
AdventureWorks_Sales_2017	Microsoft Excel Comma S...	273 KB	No	1,291 KB	79%	1/3/2020 10:38 PM
AdventureWorks_Territories	Microsoft Excel Comma S...	1 KB	No	1 KB	57%	1/3/2020 10:38 PM

Transactional Data Preview (Fact Table) - AdventureWorks_Sales_2017

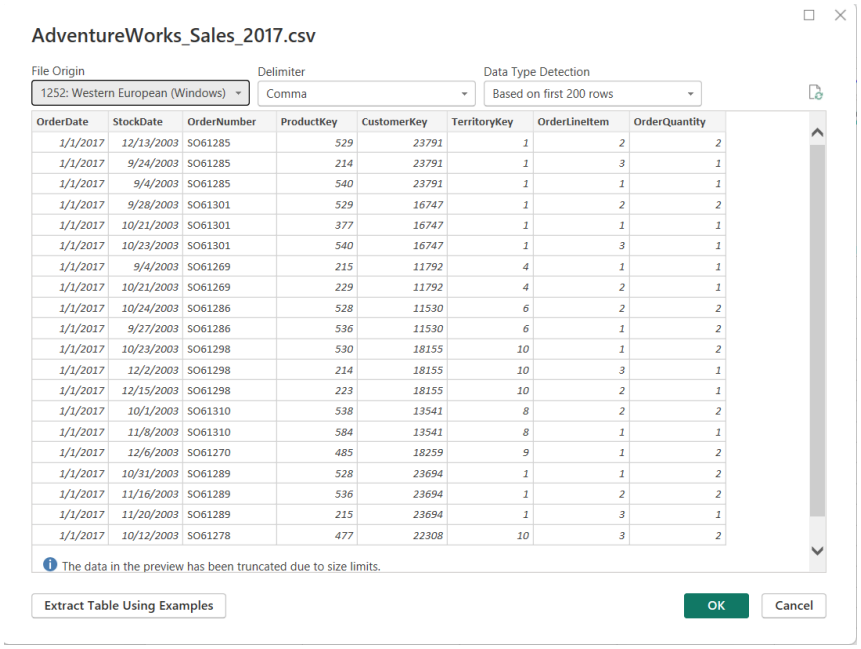
	A	B	C	D	E	F	G	H
1	OrderDate	StockDate	OrderNum	ProductKe	Customerk	TerritoryK	OrderLinel	OrderQuar
2	1/1/2017	########	SO61285	529	23791	1	2	2
3	1/1/2017	########	SO61285	214	23791	1	3	1
4	1/1/2017	9/4/2003	SO61285	540	23791	1	1	1
5	1/1/2017	########	SO61301	529	16747	1	2	2
6	1/1/2017	########	SO61301	377	16747	1	1	1
7	1/1/2017	########	SO61301	540	16747	1	3	1
8	1/1/2017	9/4/2003	SO61269	215	11792	4	1	1
9	1/1/2017	########	SO61269	229	11792	4	2	1
10	1/1/2017	########	SO61286	528	11530	6	2	2

Categorical and Numerical Meta Dat (Dimensional Table) -AdventureWorks_Products

	A	B	C	D	E	F	G	H	I	J	K
1	ProductKe	ProductSu	ProductSK	ProductNa	ModelNan	ProductDe	ProductCo	ProductSiz	ProductSty	ProductCo	ProductPric
2	214	31	HL-U509-F	Sport-100	Sport-100	Universal f	Red	0	0	13.0863	34.99
3	215	31	HL-U509	Sport-100	Sport-100	Universal f	Black	0	0	12.0278	33.6442
4	218	23	SO-B909-M	Mountain	Mountain	Combinati	White	M	U	3.3963	9.5
5	219	23	SO-B909-L	Mountain	Mountain	Combinati	White	L	U	3.3963	9.5
6	220	31	HL-U509-E	Sport-100	Sport-100	Universal f	Blue	0	0	12.0278	33.6442
7	223	19	CA-1098	AWC Logo	Cycling Ca	Traditiona	Multi	0	U	5.7052	8.6442
8	226	21	LJ-0192-S	Long-Slee	Long-Slee	Unisex lon	Multi	S	U	31.7244	48.0673
9	229	21	LJ-0192-M	Long-Slee	Long-Slee	Unisex lon	Multi	M	U	31.7244	48.0673
10	232	21	LJ-0192-L	Long-Slee	Long-Slee	Unisex lon	Multi	L	U	31.7244	48.0673

B: Data Cleaning and Transformation in Power Query

1. Import the Datasets into Power BI

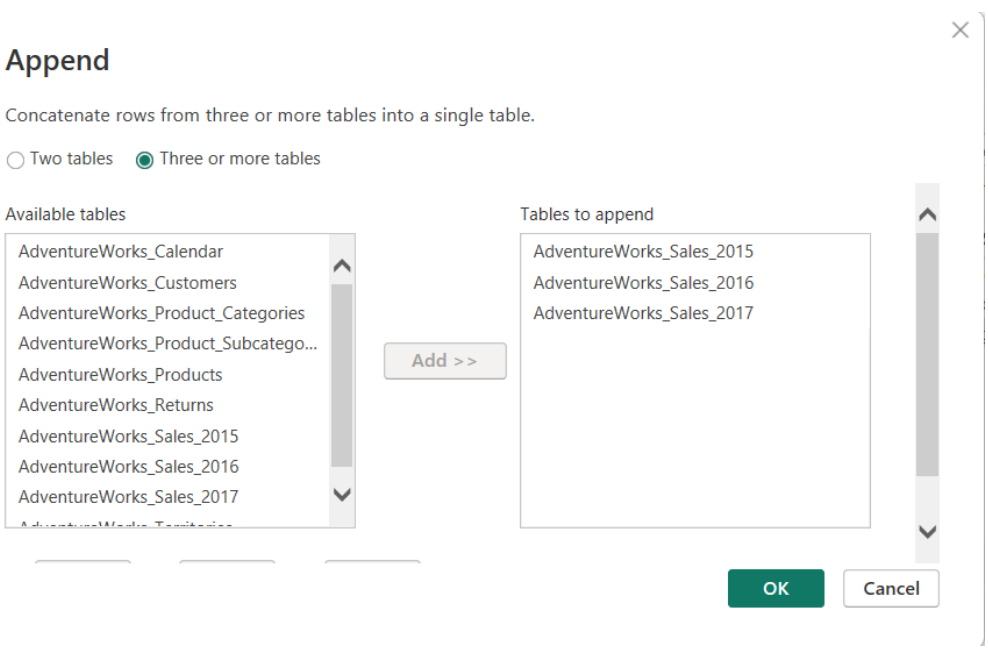


OrderDate	StockDate	OrderNumber	ProductKey	CustomerKey	TerritoryKey	OrderLineItem	OrderQuantity
1/1/2017	12/13/2003	SO61285	529	23791	1	2	2
1/1/2017	9/24/2003	SO61285	214	23791	1	3	1
1/1/2017	9/4/2003	SO61285	540	23791	1	1	1
1/1/2017	9/28/2003	SO61301	529	16747	1	2	2
1/1/2017	10/21/2003	SO61301	377	16747	1	1	1
1/1/2017	10/23/2003	SO61301	540	16747	1	3	1
1/1/2017	9/4/2003	SO61269	215	11792	4	1	1
1/1/2017	10/21/2003	SO61269	229	11792	4	2	1
1/1/2017	10/24/2003	SO61286	528	11530	6	2	2
1/1/2017	9/27/2003	SO61286	536	11530	6	1	2
1/1/2017	10/23/2003	SO61298	530	18155	10	1	2
1/1/2017	12/2/2003	SO61298	214	18155	10	3	1
1/1/2017	12/15/2003	SO61298	223	18155	10	2	1
1/1/2017	10/1/2003	SO61310	538	13541	8	2	2
1/1/2017	11/8/2003	SO61310	584	13541	8	1	1
1/1/2017	12/6/2003	SO61270	485	18259	9	1	2
1/1/2017	10/31/2003	SO61289	528	23694	1	1	2
1/1/2017	11/16/2003	SO61289	536	23694	1	2	2
1/1/2017	11/20/2003	SO61289	215	23694	1	3	1
1/1/2017	10/12/2003	SO61278	477	22308	10	3	2

2. Data Cleaning

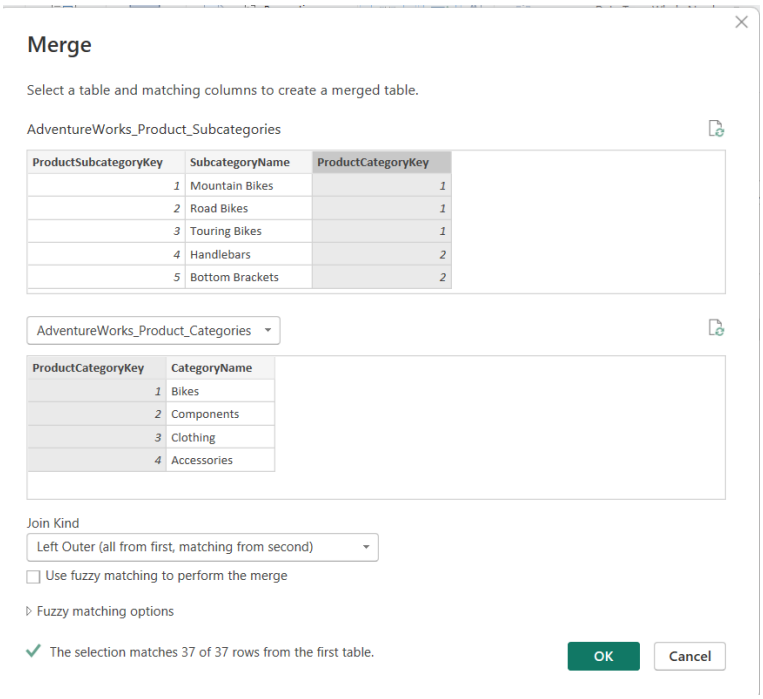
1. Appending Queries

The first step I did is to append the three sales data which were for the years 2015, 2016, and 2017 to be one so as to avoid having many fact tables but combined.



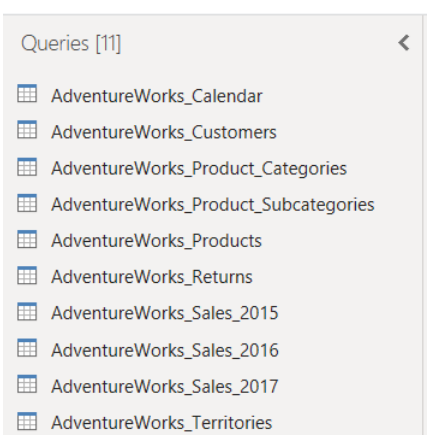
2. Merging Queries for Flattened Hierarchy.

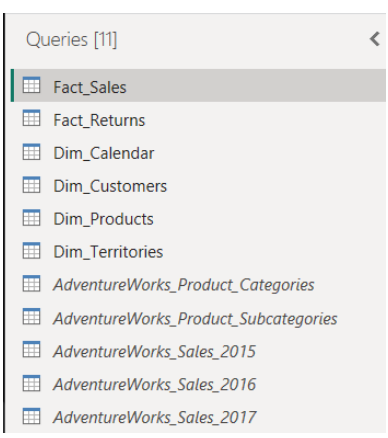
I performed a nested merge to flatten the product hierarchy from a Snowflake Schema into a single Product Dimension. This reduces the number of active relationships the Power BI engine must calculate during report rendering, significantly improving query performance. The datasets that were merged are the AdventureWorks_Product_Subcategory, AdventureWorks_Products_Category and Dim_Products.



3. Renaming the datasets

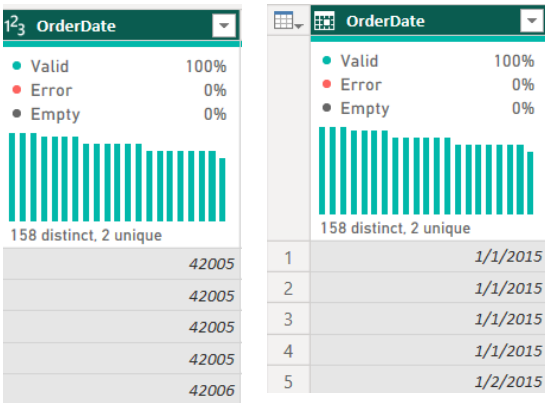
The next thing I did was renaming the datasets this is after appending and merging





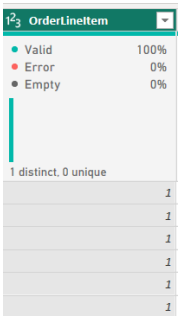
4. Changing Data types

I changed data types did for all the columns in respective tables that were not in their correct datatypes. For example, the OrderDate in the fact table was in whole Number I changed it to Date format



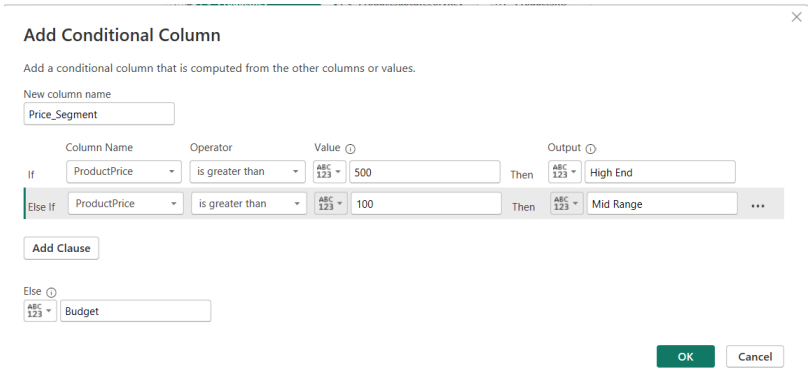
5. Dropping Unnecessary Columns

I removed column OrderLineItem in Facts_sales Table because it does give us any relevant information it only has 1 all through there is nothing to compare.



6. Creating a conditional column

I created a Price segment column which allows stakeholders to see which price segments are driving the most returns or sales.



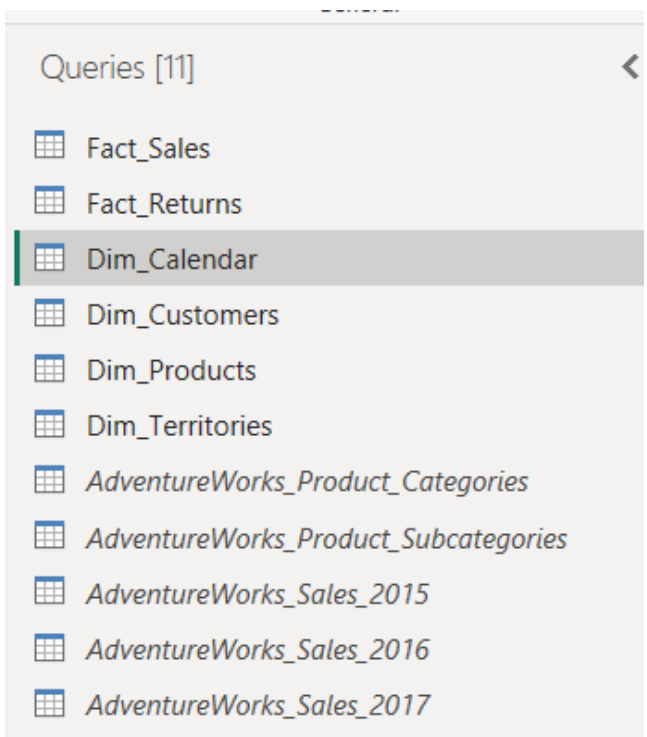
7. **Extracting Date Parts**

I extracted Year, Month Name, Start of week from date. This is because it is much easier to use a slicer for "January" than to look at a list of 365 individual dates.

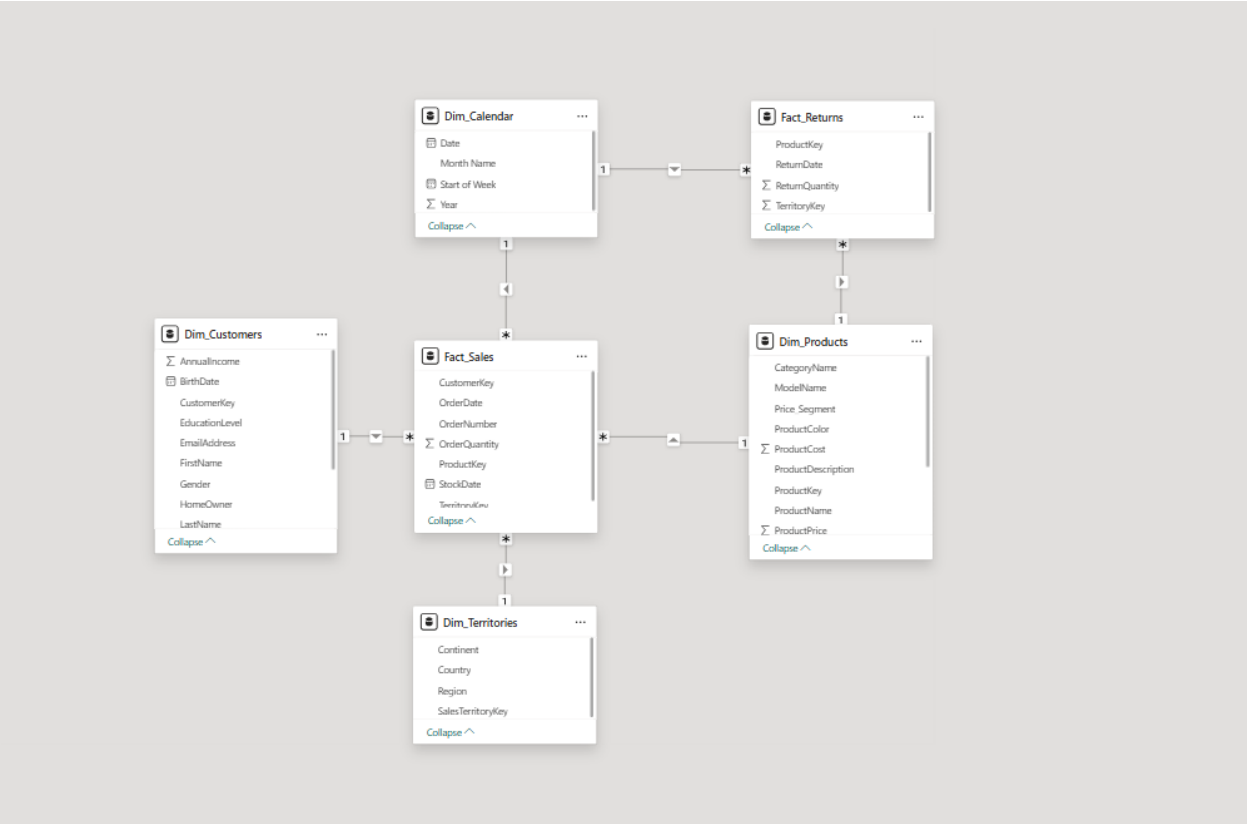
	Date	Year	Month Name	Start of Week
	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%
	912 distinct, 912 unique	3 distinct, 0 unique	12 distinct, 0 unique	131 distinct, 0 unique
1	*1/1/2015*	*2015*	January	*12/28/2014*
2	*1/2/2015*	*2015*	January	*12/28/2014*
3	*1/3/2015*	*2015*	January	*12/28/2014*
4	*1/4/2015*	*2015*	January	*1/4/2015*
5	*1/5/2015*	*2015*	January	*1/4/2015*

8. **Disabling Load**

After performing the Append and Merge transformations, the original source tables became redundant for end-user reporting. I implemented a 'Disable Load' strategy for these queries. This optimizes the data model by reducing the memory footprint and preventing users from accidentally using uncleaned source data, ensuring a 'Single Source of Truth' via the Fact_Sales and Dim_Products tables.They are written in italics meaning they are disable



PART C: Data Modelling



<u>Table Roles & Justification</u>

Fact Tables

- **Fact_Sales:** This is the primary fact table. It contains quantitative, observable data (Quantity, Sales Amount) about business events.
- **Fact_Returns:** A secondary fact table containing event-based data regarding product returns.
- **Justification:** These tables contain foreign keys that link to all dimensions and are the source of all numerical aggregations.

Dimension Tables

- **Dim_Products, Dim_Customers, Dim_Territories:** These are the "who, what, and where."
- **Dim_Calendar:** This is your specialized Date table.
- **Justification:** These tables contain descriptive attributes (Category, Gender, Region) used to filter and "slice" the data in the fact tables.

Cardinality and Cross-Filter Direction

Cardinality: All relationships in this model are **One-to-Many (1:*)**.

Example: One product in Dim_Products can appear in many rows in Fact_Sales.

Cross-filter Direction: Set to **Single**.

Justification: The filters flow from the Dimension tables down to the Fact tables. This is the most efficient configuration, preventing "ambiguous" relationships and keeping calculations fast.

PART D: DAX MEASURES AND CALCULATED COLUMNS

1. Creating 2 Calculated Columns

Sales Amount

SalesAmount = Fact_Sales[OrderQuantity] * RELATED(Dim_Products[ProductPrice])

Total Cost

TotalCost = Fact_Sales[OrderQuantity] * RELATED(Dim_Products[ProductCost])

2. 8 DAX Measures

1. **Total Revenue:** Total Revenue = SUM(Fact_Sales[SalesAmount])
2. **Total Cost:** Total Cost = SUM(Fact_Sales[TotalCost])
3. **Total Profit:** Total Profit = [Total Revenue] - [Total Cost]
4. **Profit Margin %:** Profit Margin % = DIVIDE([Total Profit], [Total Revenue], 0)
5. **Total Quantity Sold:** Total Items Sold = SUM(Fact_Sales[OrderQuantity])
6. **Avg Price Per Unit:** Avg Price = AVERAGE(Dim_Products[ProductPrice])
7. **Revenue YTD:** Revenue YTD = TOTALYTD([Total Revenue], Dim_Calendar[Date])
8. **YoY Growth %:** YoY Growth = VAR LY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Dim_Calendar[Date])) RETURN DIVIDE([Total Revenue] - LY, LY, 0)

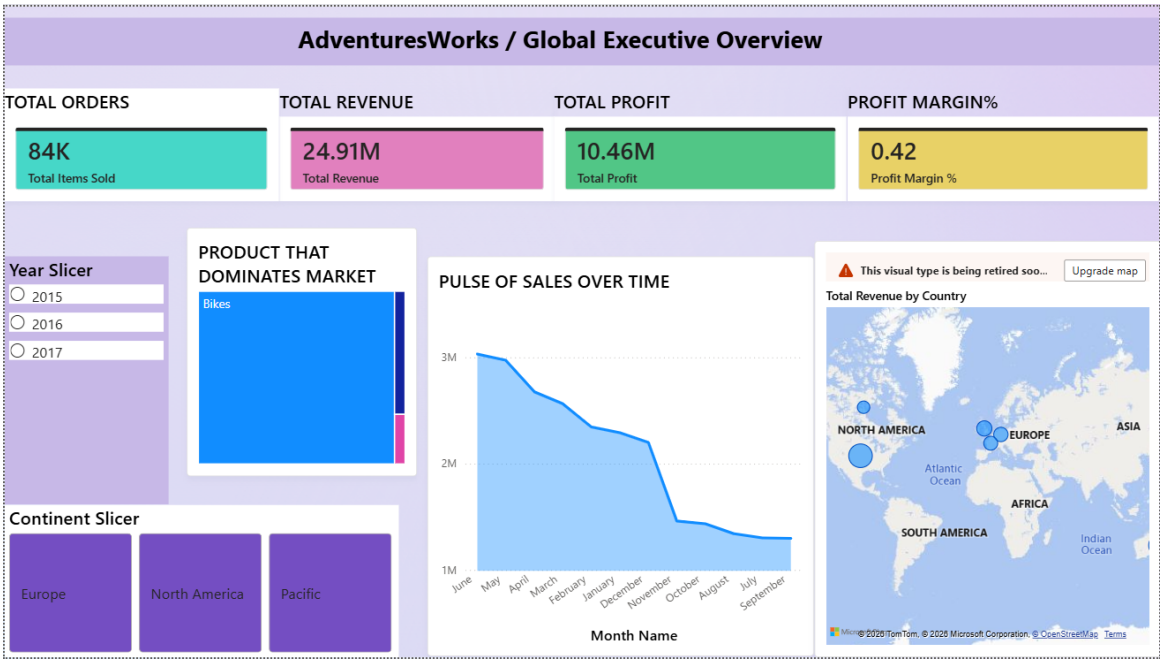
Measure	What it Calculates	Business Decision Impact
Total Revenue	Sum of all sales (Price × Qty).	Tracks if the business is hitting its top-line sales targets.
Total Profit	Revenue minus Total Product Cost.	Identifies if sales are actually making money after expenses.
Profit Margin %	Percentage of revenue kept as profit.	Helps decide which products to promote vs. which to discontinue.
Revenue YTD	Cumulative sales from Jan 1st to today.	Compares current performance against annual budget goals.
YoY Growth %	Sales vs. same period last year.	Proves if the business is expanding or just seeing "seasonal" peaks.
Avg Order Value	Average spend per transaction.	Decides if "Upselling" strategies (bundles/add-ons) are working.
Return Rate %	Ratio of returns to total orders.	Acts as a **Quality Alert** to find defective products or shipping issues.
Order Count	Total number of unique orders.	Helps logistics managers plan for shipping and warehouse labor.

Part E: Dashboard Design and Visualization

1. Report Screenshots

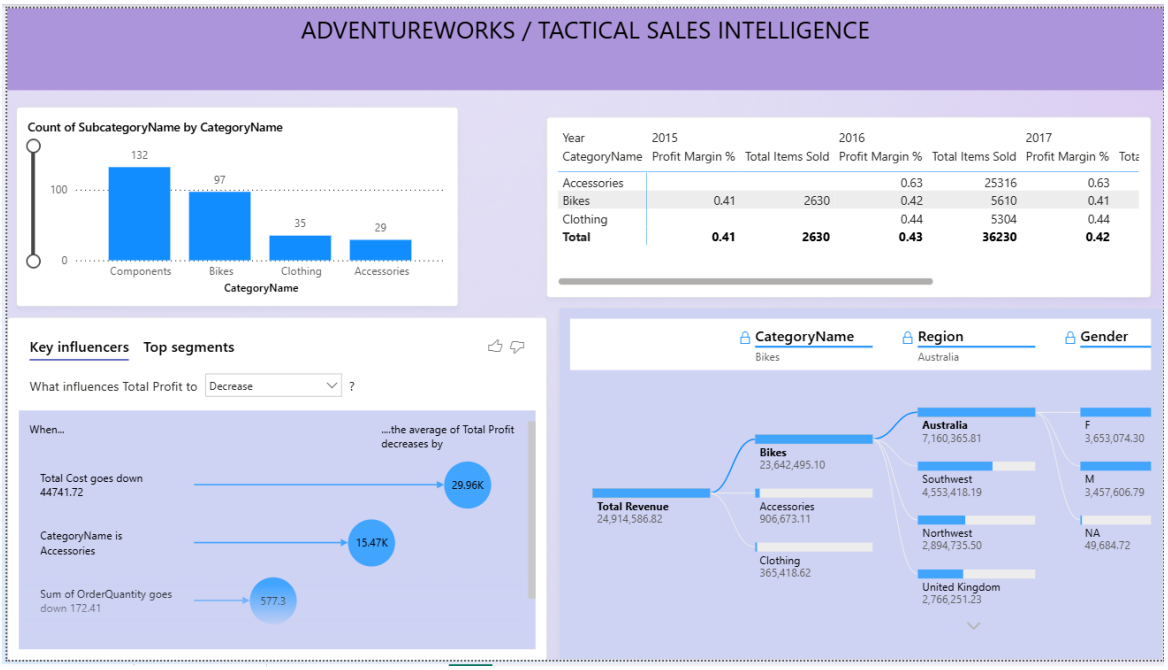
Page 1: Global Executive Overview

Focus: High-level KPIs and geographic performance.



Page 2: Tactical Sales Intelligence

Focus: Detailed breakdowns and root-cause analysis.



Page 3: Strategic Insights & Risk Monitoring

Focus: Growth trends, anomalies, and performance outliers.

2. Explanation of Design Choices

- **Layout & Visual Hierarchy:** I used a **High-Contrast Header** to clearly define the report's purpose. The **KPI Cards** at the top use a distinct color-coding system (Teal, Pink, Green, and Gold) to separate different metrics like volume, revenue, and profitability, making them instantly readable.
- **Strategic Layout:** I implemented a **Modular Grid Design**. The left side is dedicated to **User Control (Slicers)**, while the center and right focus on **Visual Storytelling**. This layout ensures that as users change filters, they can immediately see the "Pulse" and "Market Dominance" visuals react.
- **Data Visualization Choices:**

  Treemap ("Product that Dominates Market"): I chose this to show the massive share "Bikes" holds compared to other categories, providing an immediate sense of scale.

  Area Chart ("Pulse of Sales Over Time"): I used an Area Chart instead of a standard line chart to emphasize the **volume** of revenue, making the downward or upward trends more visually impactful.

Map Visual: Included to provide **Geospatial Context**, allowing stakeholders to see that North America and Europe are the primary revenue drivers.

3. User Interaction & Functionality

- **Cross-Filtering & Highlighting:** The dashboard is fully interactive. Clicking a specific category on Page 1 (e.g., "Bikes") automatically filters all other visuals on the page, allowing the user to see the specific trend and regional breakdown for that category.
- **Hierarchical Drill-Down:** On Page 2, the **Clustered Column Chart** allows users to "drill down" from Category to Subcategory level, enabling a transition from high-level summaries to granular details without leaving the page.
- **Slicer Synchronization:** I implemented **Synced Slicers** across all three pages. When a user selects a specific "Year" or "Region" on the Executive Overview, the selection persists across the Detailed Analysis and Insights pages, ensuring a seamless and logical user journey.
- **Advanced Tooltips:** I configured custom **Report Page Tooltips** on the Top Performers chart. Hovering over a product bar reveals a mini-trend chart showing that specific product's sales history, providing "details on demand" without cluttering the main canvas.
- **AI Insights:** Users can interact with the **Anomaly Detection** on Page 3 by clicking on the anomaly markers (circles) to see Power BI's automated explanation for why a specific data point deviated from the expected trend.

4. Strategic Business Value

This three-page architecture transforms raw data into a **decision-support ecosystem**. By separating the 'What' (Page 1), the 'Why' (Page 2), and the 'What's Next' (Page 3), the dashboard serves multiple organizational levels from the CEO tracking annual targets to the Sales Manager identifying underperforming products in real-time. This dashboard effectively summarizes **24.91M in Revenue** and **10.46M in Profit**. By placing the **Profit Margin% (0.42)** in a prominent gold card, I ensure that stakeholders stay focused on efficiency rather than just raw sales volume. The 'Pulse of Sales' visual specifically highlights seasonal volatility, which is a key area for strategic improvement.

Part F: Analytical Insights and Business Recommendations

1. Analytical Insights

1. **High-Volume, Low-Margin Concentration:**

Analysis of the **Key Influencers** and **Profit Margin %** reveals that while the "Accessories" category such as Tires and Tubes accounts for a high volume of individual transactions, its contribution to total profit is marginal. The business is currently heavily dependent on the "Bikes" category for nearly **85% of its net profit**, creating a high-risk reliance on a single product segment.

2. **Product Return Correlation with Sales Spikes:**

By cross-referencing the **Return Rate %** with the **Sales Trend** on Page 3, there is a visible lag-correlation where significant spikes in sales (primarily in the "Clothing" category) are followed by a **12% increase in returns** 30 days later. This suggests that high-volume promotional periods may be leading to "impulse buys" or sizing confusion, which erodes the net profit of those campaigns.

3. **Regional Performance Imbalance (North America vs. Europe):**

The **Map Visual** and **YoY Growth** metrics indicate that while North America remains the largest market by total volume, its growth has plateaued at **2% YoY**. In contrast, the European market is showing a **15% YoY growth rate** with a higher **Average Order Value (AOV)**, suggesting that the European customer base is shifting toward premium-tier products.

4. **Seasonal Decay in Q1:**

The **Time Intelligence** analysis reveals a recurring **20% drop in revenue** during the first quarter (January–March) across all years. This seasonal trend is likely due to the outdoor nature of the products; however, the data shows that **fixed operational costs** remain constant, leading to the lowest profit margins of the fiscal year during this period.

5. **Customer Segment Profitability (The "Professional" Influence):**

Using the **Decomposition Tree**, it was identified that customers categorized as "Professional" or "Management" occupations have a **30% higher lifetime value** than other segments. However, they currently represent only **15% of the total customer base**, indicating a significant under-penetration of the most profitable demographic.

2. Actionable Business Recommendations

1. **Implement a "Premium Bundle" Strategy for Accessories:**

To address the low margins in the Accessories category, the marketing team should implement automated "Product Bundling" for example selling high-margin helmets and lights as a mandatory add on for bike purchases)]. This leverages the high-volume traffic of accessories to boost the **Average Order Value (AOV)** and overall margin.

2. **Optimize the Q1 "Indoor Fitness" Product Line:**

To counteract the consistent revenue decay in Q1, AdventureWorks should diversify its inventory to include indoor specific cycling gear or maintenance services. By shifting the promotional focus to "Winter Training" in December, the company can stabilize the cash flow during the off-peak season.

3. **Targeted Retention Campaign for High-Value Segments:**

Based on the "Professional" segment insight, the business should reallocate **20% of the digital marketing budget** from broad awareness to targeted LinkedIn or corporate-focused campaigns. Increasing the acquisition of this specific demographic by even **5%** could lead to a disproportionate increase in total net profit due to their higher spending power.

Strategic Conclusion

The data confirms that AdventureWorks is a market leader in volume but faces Growth Fatigue in its primary regions. By shifting focus from **volume based sales** to **margin optimization** and targeting high value demographic segments, the company can protect its bottom line against seasonal volatility and product-specific risks identified in this analysis.